

ClimateScope

Milestone – 3 : Visualization Development & Interactivity

1. Introduction

Milestone 3 of the ClimateScope project focuses on converting processed climate data into meaningful, interactive, and visually compelling dashboards. The primary objective of this milestone is to enable users to explore climate patterns, trends, correlations, and extreme events through both static and interactive visualizations. By leveraging modern visualization tools and interactive UI components, this milestone bridges the gap between raw analysis and actionable insights.

2. Visualization Tools and Technologies Used

2.1 Selected Visualization Tools

For this milestone, the following tools were used:

- **Plotly:** For interactive charts such as line plots, area charts, scatter plots, histograms, heatmaps, and geographic maps.
- **Streamlit:** For building the interactive dashboard interface, managing user inputs, and enabling real-time visualization updates.

These tools were selected due to their strong interactivity, ease of integration with Python, and support for professional-grade visual analytics.

3. Development of Static and Interactive Visualizations

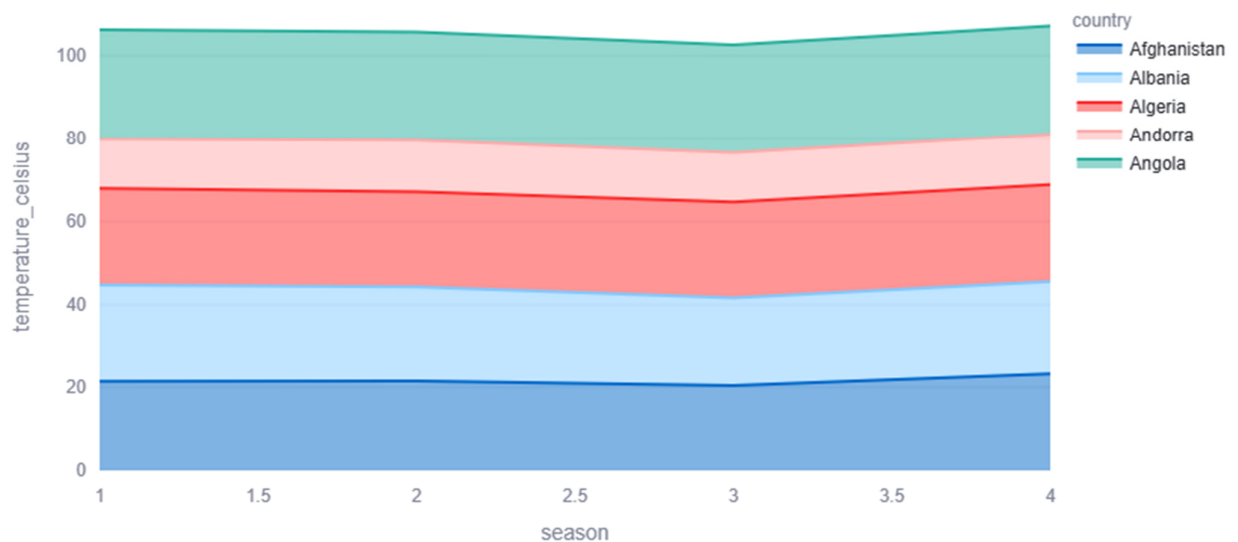
3.1 Trend Analysis Visualizations

To analyze long-term climate behavior, time-series visualizations were developed:

- **Line Charts** display changes in climate parameters (temperature, humidity, wind speed, precipitation) over time.
- **Area Charts** highlight cumulative and intensity-based trends.

Insight:

These plots reveal gradual warming patterns, seasonal fluctuations, and long-term variability across regions.



3.1.1 Comparing temperature_celsius between selected countries

3.2 Distribution Analysis Visualizations

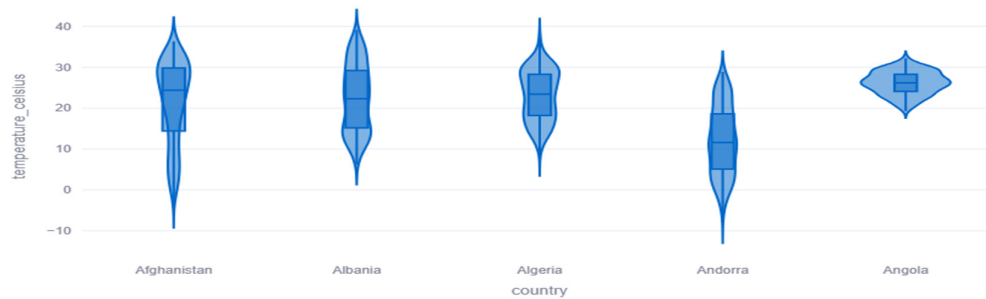
To understand data spread and variability, the following plots were implemented:

- **Histogram** for frequency distribution of climate variables.
- **Box Plot** to detect outliers and median variations.
- **Violin Plot** to show distribution density and skewness.
- **Density Plot** for smooth probability distribution visualization.

Insight:

Distribution plots help identify extreme values, variability across countries, and parameter stability.

3.2.1 Violin Plot



3.2.2 Box Plot

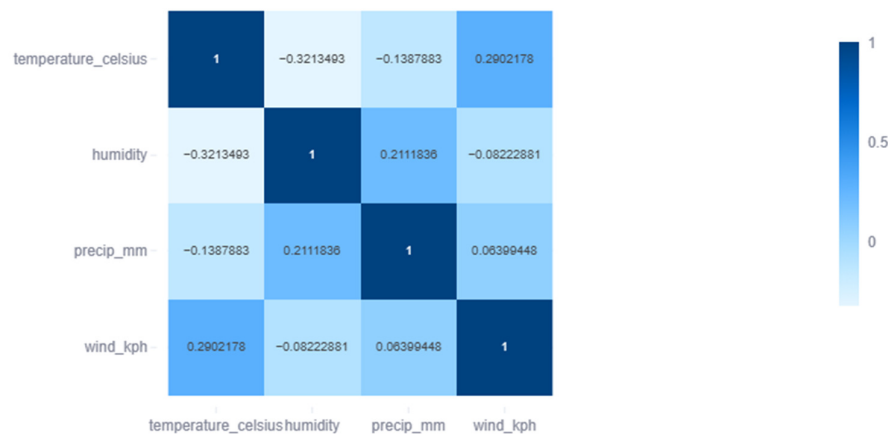
3.3 Correlation Analysis Visualizations

To study relationships between climate variables:

- **Scatter Plots** show pairwise relationships.
- **Bubble Charts** incorporate a third variable using size encoding.
- **Correlation Heatmap** quantifies the strength and direction of relationships.

Insight:

Strong correlations were observed between humidity and precipitation, while temperature showed varying relationships with wind speed.



3.4 Seasonal Analysis Visualizations

Seasonal trends were explored using:

- **Monthly Aggregated Line Charts**
- **Monthly Bar Charts**
- **Seasonal Heatmaps**
- **Ridgeline (Joy) Plots**

Insight:

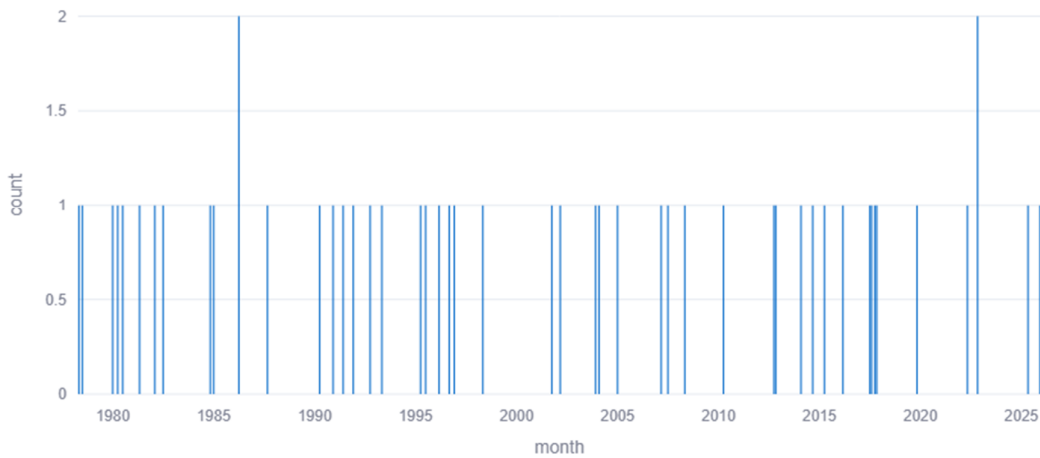
These plots clearly expose recurring seasonal patterns and climate cyclicity across different months and regions.

3.4.1 Line Chart

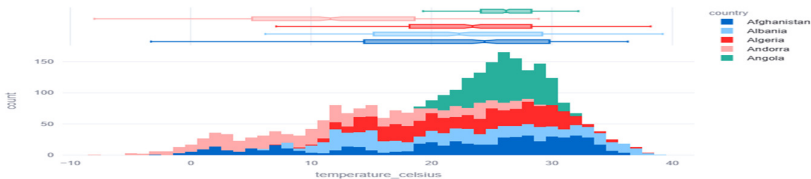
Metric Trend



3.4.2 Thin - Bar Chart



3.4.3 Stacked Histogram



3.5 Extreme Events Analysis Visualizations

Extreme climate events were identified using user-defined thresholds:

- **Top-Five Tables** for hottest, coldest, wettest, and windiest days.
- **Frequency Bar Charts** showing extreme event counts.
- **Monthly Extreme Event Trend Line Charts**
- **Regional Extreme Events Tables**

Insight:

The analysis highlights increasing frequency of extreme events in certain regions and seasons.

Top 5 Extreme Events

	country	last_updated	temperature_celsius
14110	Albania	2017-08-31 00:00:00	39.2
7194	Albania	1986-04-24 00:00:00	38.2
21909	Algeria	1991-06-12 00:00:00	38.2
12575	Albania	2017-07-01 00:00:00	38.1
17229	Albania	2012-11-28 00:00:00	38.1



4. Integration of User Interactivity

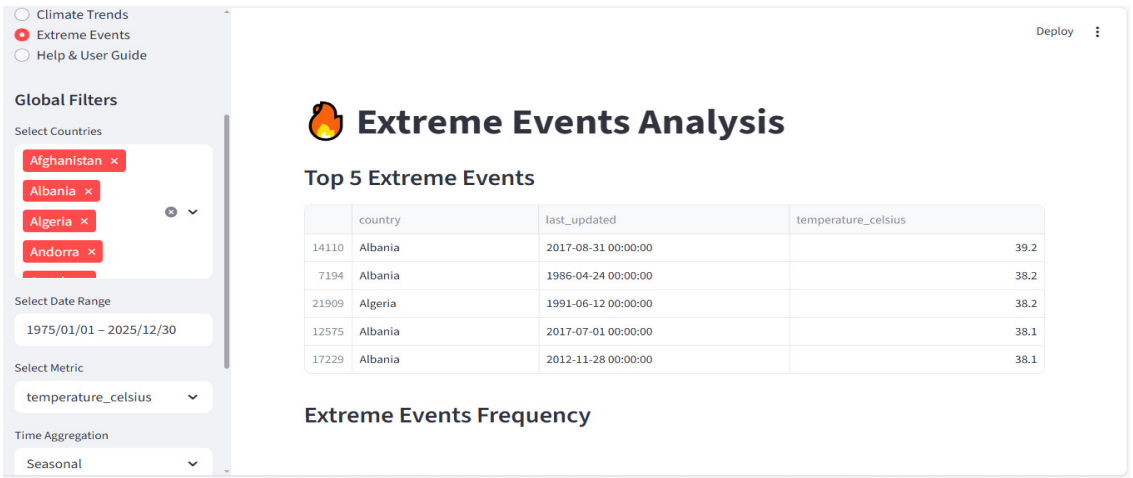
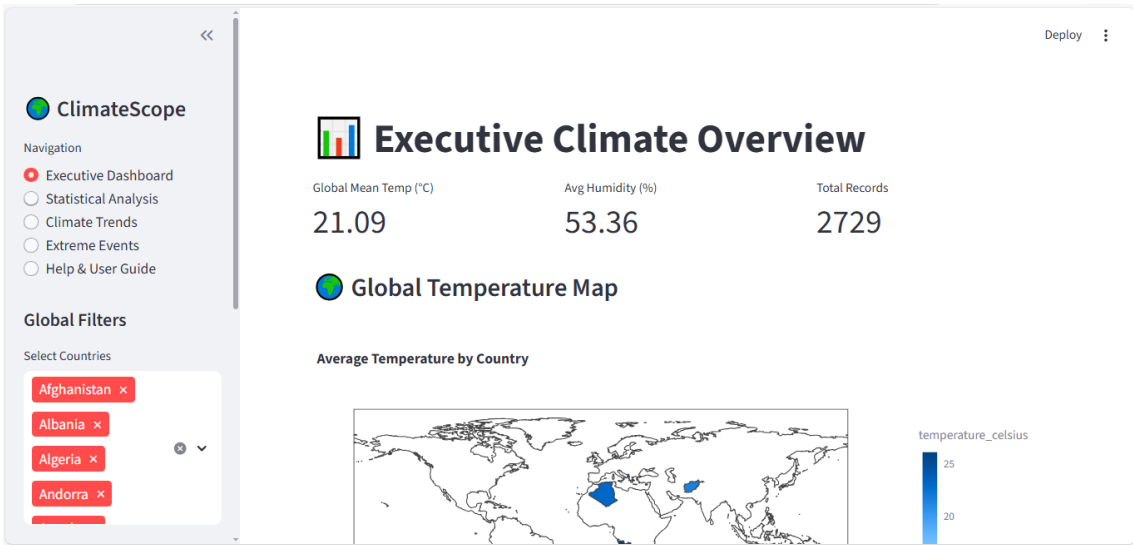
4.1 Filters and Selectors

The dashboard includes:

- **Country Multi-Select Filter**
- **Date Range Selector (Dataset-Driven)**
- **Metric Selector**
- **Threshold Slider for Extreme Events**
- **Time Aggregation Selector (Daily, Monthly, Seasonal)**
- **Normalization Toggle**

These controls dynamically update all visualizations in real time.

Dashboard View



5. Key Findings and Insights from the Dashboard

- Long-term warming trends are observable across multiple regions.
- Seasonal variability significantly impacts temperature and precipitation.
- Strong correlations exist between humidity and rainfall.
- Extreme weather events show increasing frequency in specific regions.
- Geographic visualization reveals spatial clustering of climate risks.

6. Limitations and Future Enhancements

6.1 Current Limitations

- Dependence on historical data
- Absence of real-time weather feeds
- Limited predictive capabilities

6.2 Future Scope

- Integration of live weather APIs
- Machine learning-based trend prediction
- Deployment as a public web application

7. Conclusion

Milestone 3 successfully delivers a fully interactive, visually rich climate dashboard that transforms complex climate data into actionable insights. The implementation of advanced visualizations, user interactivity, and refined UI design fulfills all milestone objectives and establishes a strong foundation for further analytical and predictive development in the ClimateScope project.